Status	Finished
Started	Sunday, 25 October 2026, 6:08 PM
Completed	Sunday, 25 October 2026, 6:16 PM
Duration	7 mins 43 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Determining whether an element is present in a list ✓
- b. Sorting elements in a list
- c. Deleting elements from a list
- d. Inserting elements into a list

Question **2**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the left sublist
- b. At the beginning of the list
- c. In the right sublist ✓
- d. In the middle of the list

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It does not work with negative numbers
- b. It is slower than linear search for small lists
- c. It requires the list to be sorted ✓
- d. It can only be applied to large lists

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly

b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

c. Must use a sorted array

d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements

b. When the list is sorted ✓

c. When the list is unsorted

d. When the list is very small

Question **6**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)

b. O👎

c. O(n log n)

d. O(log n) ✓

Question **7**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1

b. n ✓

c. log n

d. n/2

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The search continues from the beginning
- c. The search stops
- d. The list is sorted

Question **9**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the entire list
- b. In the left sublist ✓
- c. At the beginning of the list
- d. In the right sublist

Question **10**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search continues until the end of the list
- b. The search backtracks to find duplicate elements
- c. The search starts over from the beginning
- d. The search stops immediately ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By sorting the list first
- b. By using a hash function
- c. By comparing each element in the list sequentially ✓
- d. By dividing the list into halves

Question **12**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 99 ✓

b. 90 and 100

c. 94 and 99

d. 89 and 94

Question **13**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Divide the list into two equal parts

b. Compare the target element with the middle element in the list ✓

c. Sort the list

d. Compare the target element with the first element in the list

Question **14**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)

b. Hash search

c. Linear search ✓

d. Binary search

Question **15**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Insertion

c. Bubble ✓

d. Complexity